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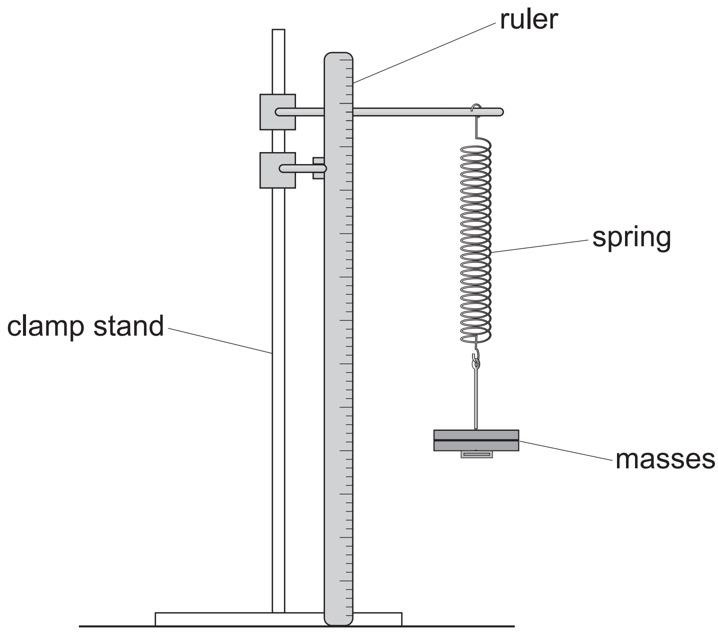
5. Some students carry out an experiment to determine the relationship between the force applied to a spring and its extension.
They use the following apparatus.



They collect data by loading and then unloading the spring. Their results are shown in the table below.

Mass (g)	Force (N)	Extension (cm)	
		Loading	Unloading
0	0	0.0	0.0
100	1	4.4	4.2
200	2	8.9	9.3
300	3	13.6	13.7
400	4	18.2	18.5
500	5	22.5	22.5

(a) The uncertainty in extension is given by:

$$\text{uncertainty in extension} = \frac{\text{maximum value of extension} - \text{minimum value of extension}}{2}$$

(i) State the mass with the greatest uncertainty in extension. [1]

value of mass = ... g

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(ii) Calculate this uncertainty. [2]

uncertainty = ... cm

(b) Suggest how the students could adapt the apparatus to reduce the uncertainty in their results. [1]

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(c) Explain how they could use **all** of the results to determine a mean value for the spring constant. [3]

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(d) The students determine the spring constant to be 22.3 N/m.
The true value of the spring constant is 22.2 N/m.
Explain what this tells the students about their data. [2]

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